ST7 - TD 6: Binomial Model (Detailed Corrections)

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Exercise 1: Pricing of barrier option

Context and Definitions

We are evaluating an **up-and-out call option**. The payoff is given by:

$$H := \mathbf{1}_{\{M_T \leq B\}}(S_T - K)^+$$

where $(M_t)_{0 \leq t \leq T}$ is the running maximum process $M_t := \max_{0 \leq u \leq t} S_u$, $B > 1$ is the barrier, and $K > 0$ is the strike price. If the asset price ever exceeds B, the option becomes worthless.

Question 1: Arbitrage-free, completeness, and martingale measure

Goal: Show the market is arbitrage-free and complete, and specify the equivalent martingale measure Q.

Step-by-step Solution:

1. **Arbitrage-free condition:** By the Fundamental Theorem of Asset Pricing, a market is arbitrage-free if and only if there exists an equivalent martingale measure (EMM) Q. For a single-period binomial model, this implies the risk-free rate must lie strictly between the downward and upward return factors. Thus, the condition is $-1 < d < r < u$.

2. **Completeness:** Because there are exactly two possible states for the asset return at each time step (up or down) and two assets (the risky asset and the risk-free bank account), the market is complete. Any derivative can be perfectly replicated.

3. **Specifying Q:** Under the EMM Q, the discounted asset price must be a martingale. This means the expected return of the risky asset under Q equals the risk-free rate r:

   $$E_Q[R_t] = r \implies u \cdot Q[R_t = u] + d \cdot Q[R_t = d] = r$$

   Using the fact that probabilities sum to 1 ($Q[R_t = d] = 1 - Q[R_t = u]$):

   $$u \cdot q + d \cdot (1 - q) = r \implies q(u - d) = r - d \implies q = \frac{r - d}{u - d}$$

   Thus, under Q, $R_1, \ldots, R_T$ are independent and identically distributed (i.i.d.) random variables with $Q[R_t = u] = q$ and $Q[R_t = d] = 1 - q$.

Question 2: The Stopping Time

Goal: Let $\tau := \inf\{0 \leq t \leq T : S_t > B\}$. Show it is a stopping time and $\tau = \inf\{0 \leq t \leq T : M_t > B\}$.

Step-by-step Solution:

1. **Proving it is a stopping time:** By definition, a random variable τ is a stopping time if the event $\{\tau \leq t\}$ belongs to the filtration $\mathcal{F}_t$ for all t. Equivalently, we can check the complement $\{\tau > t\}$.

$$\{\tau > t\} = \{S_0 \leq B, S_1 \leq B, \ldots, S_t \leq B\}$$

Since this event only depends on the asset prices up to time t, it is $\mathcal{F}_t$-measurable. Thus, τ is a stopping time.

2. **Equivalence with the running maximum:** We evaluate the two possible scenarios for the stopping time:

   - **Case 1 ($\tau = +\infty$):** The barrier is never breached.

   $$S_t \leq B, \forall 0 \leq t \leq T \iff \max_{0 \leq t \leq T} S_t \leq B \iff M_T \leq B$$

   In this case, the set $\{0 \leq t \leq T : M_t > B\}$ is empty, so its infimum is $+\infty$.

   - **Case 2 ($\tau \leq T$):** The barrier is breached at some specific time τ.

   $$S_t \leq B, \forall 0 \leq t \leq \tau - 1 \quad \text{and} \quad S_\tau > B$$

   This implies the running maximum up to $\tau - 1$ is below the barrier ($M_{\tau-1} \leq B$), but at time τ, the maximum exceeds the barrier ($M_\tau > B$). Therefore, the first time $S_t > B$ is identical to the first time $M_t > B$.

Question 3: Recursive Pricing Formula

Goal: Prove the value of the option $v_t := E_Q[H/(1+r)^{T-t} \mid \mathcal{F}_t]$ can be written as $v_t = f_t(S_t, M_t)$ and find the recursive relationship.

Step-by-step Solution:

1. **Terminal Condition ($t = T$):** At maturity, the value of the option is simply its payoff.

$$v_T = H = \mathbf{1}_{\{M_T \leq B\}}(S_T - K)^+$$

This strictly depends on S_T and M_T, giving us our base function: $f_T(s, m) := \mathbf{1}_{\{m \leq B\}}(s - K)^+$.

2. **Backward Induction Step ($t-1$):** We want to express v_{t-1} in terms of v_t. Using the Tower Property of conditional expectations:

$$v_{t-1} = \frac{1}{1+r} E_Q[v_t \mid \mathcal{F}_{t-1}]$$

We split the expectation based on whether the barrier was breached *before or at* time $t-1$:

$$v_{t-1} = \frac{1}{1+r} E_Q[(\mathbf{1}_{\{M_{t-1} \leq B\}} + \mathbf{1}_{\{M_{t-1} > B\}}) v_t \mid \mathcal{F}_{t-1}]$$

If $M_{t-1} > B$, the barrier is already breached, meaning $M_T > B$ is guaranteed, and the final payoff $H = 0$. Thus, the second term vanishes.

$$v_{t-1} = \frac{\mathbf{1}_{\{M_{t-1} \leq B\}}}{1+r} E_Q[v_t \mid \mathcal{F}_{t-1}]$$

3. **Applying the Markov Property:** Assuming $v_t = f_t(S_t, M_t)$ holds for t, we substitute it into the expectation. Note that $S_t = S_{t-1}(1 + R_t)$ and $M_t = \max(M_{t-1}, S_t)$.

$$v_{t-1} = \frac{\mathbf{1}_{\{M_{t-1} \leq B\}}}{1+r} E_Q[f_t(S_{t-1}(1+R_t), \max(M_{t-1}, S_{t-1}(1+R_t))) \mid \mathcal{F}_{t-1}]$$

Since S_{t-1} and M_{t-1} are known at time $t-1$ ($\mathcal{F}_{t-1}$-measurable), and R_t is independent of $\mathcal{F}_{t-1}$ under Q, we calculate the expectation over the two possible states of R_t:

- State Up: $R_t = u$ with probability q.
- State Down: $R_t = d$ with probability $1 - q$.

This directly yields the recursive formula:

$$f_{t-1}(s,m) := \frac{\mathbf{1}_{\{m \le B\}}}{1+r}\Big[(1-q)f_t\big(s(1+d), m \vee s(1+d)\big) + q f_t\big(s(1+u), m \vee s(1+u)\big)\Big]$$

(Note: $a \vee b$ is standard notation for $\max(a,b)$).

Question 4: Price Comparison

Goal: Compare the price of the barrier option H to a standard call option.

Step-by-step Solution: The standard call option has a payoff of $(S_T - K)^+$. The up-and-out call option has a payoff of $\mathbf{1}_{\{M_T \le B\}}(S_T - K)^+$. Since the indicator function $\mathbf{1}_{\{M_T \le B\}}$ can only be 0 or 1, it is strictly true that:

$$\mathbf{1}_{\{M_T \le B\}}(S_T - K)^+ \le (S_T - K)^+ \implies H \le \text{Call Payoff}$$

Because the pricing operator (taking the discounted expectation under the risk-neutral measure Q) is linear and positive, a uniformly smaller payoff must result in a smaller or equal price. Thus, the standard call option is more expensive.

Question 5: Pricing related payoffs H' and H_0

Goal: Find the prices of $H' := \mathbf{1}_{\{M_T > B\}}(S_T - K)^+$ (up-and-in call) and $H_0 := \mathbf{1}_{\{M_T \le B\}}\mathbf{1}_{\{S_T > K\}}$ (digital barrier call).

Step-by-step Solution:

1. **Pricing H' (In-Out Parity):** We notice that an up-and-out call plus an up-and-in call exactly replicates a standard call option.

$$H + H' = \mathbf{1}_{\{M_T \le B\}}(S_T - K)^+ + \mathbf{1}_{\{M_T > B\}}(S_T - K)^+ = (S_T - K)^+$$

By the linearity of pricing, $p(K) + p'(K) = \text{Price}((S_T - K)^+)$. Therefore, the price of H' is:

$$p'(K) = \text{Price(Standard Call)} - p(K)$$

2. **Pricing H_0 (Derivative Trick):** H_0 pays exactly 1 if the barrier is not breached and the final asset price is above the strike. This is a binary/digital payoff. We relate the digital payoff to the standard call payoff using calculus. For $S_T \neq K$:

$$\frac{\partial}{\partial K}(S_T - K)^+ = \begin{cases} -1 & \text{if } S_T > K \\ 0 & \text{if } S_T < K \end{cases} \implies -\frac{\partial}{\partial K}(S_T - K)^+ = \mathbf{1}_{\{S_T > K\}}$$

We multiply both sides by the barrier condition $\mathbf{1}_{\{M_T \le B\}}$:

$$H_0 = \mathbf{1}_{\{M_T \le B\}}\mathbf{1}_{\{S_T > K\}} = -\mathbf{1}_{\{M_T \le B\}}\frac{\partial}{\partial K}(S_T - K)^+$$

Taking the discounted expectation under Q to find the price $p_0(K)$:

$$p_0(K) = \frac{1}{(1+r)^T}E_Q[H_0] = \frac{1}{(1+r)^T}E_Q\left[-\mathbf{1}_{\{M_T \le B\}}\frac{\partial}{\partial K}(S_T - K)^+\right]$$

Assuming we can swap the derivative and the expectation (valid here as the state space is finite), we pull the derivative with respect to K outside:

$$p_0(K) = -\frac{\partial}{\partial K}\left(\frac{1}{(1+r)^T} E_Q\left[\mathbf{1}_{\{M_T \leq B\}}(S_T - K)^+\right]\right) = -\frac{\partial p(K)}{\partial K}$$

In practice, you approximate this derivative using a finite difference method:

$$p_0(K) \approx \frac{p(K) - p(K + \Delta K)}{\Delta K}$$

Exercise 2: Limit of binomial model

Context

We want to find an approximation of the terminal asset price S_T when the number of periods T becomes very large. This explores the connection between the discrete binomial model and continuous-time models.

Question 1: Identifying the Law

Goal: Find the probability distribution (the law) of the random variable:

$$Y_T := \frac{\ln(S_T/(S_0(1+d)^T))}{\ln((1+u)/(1+d))}$$

under the risk-neutral measure Q.

Step-by-step Solution:

1. **Expand the Asset Price:** The price at maturity T is the initial price multiplied by the consecutive returns for each period.

$$S_T = S_0(1+R_1)(1+R_2)\dots(1+R_T) = S_0\prod_{t=1}^{T}(1+R_t)$$

2. **Apply Logarithms:** Take the natural logarithm of S_T to turn the product into a sum.

$$\ln(S_T) = \ln(S_0) + \sum_{t=1}^{T}\ln(1+R_t)$$

3. **Substitute into the Target Expression:** We want to evaluate the numerator of Y_T. Notice that $\ln(S_0(1+d)^T) = \ln(S_0) + T\ln(1+d) = \ln(S_0) + \sum_{t=1}^{T}\ln(1+d)$.

$$\begin{aligned}
\ln\left(\frac{S_T}{S_0(1+d)^T}\right) &= \ln(S_T) - \ln(S_0(1+d)^T) \\
&= \left(\ln(S_0) + \sum_{t=1}^{T}\ln(1+R_t)\right) - \left(\ln(S_0) + \sum_{t=1}^{T}\ln(1+d)\right) \\
&= \sum_{t=1}^{T}\big(\ln(1+R_t) - \ln(1+d)\big) \\
&= \sum_{t=1}^{T}\ln\left(\frac{1+R_t}{1+d}\right)
\end{aligned}$$

4. **Analyze the Summand:** Now we divide by the denominator to construct Y_T. Let us define a new random variable X_t for each period t:

$$X_t := \frac{\ln((1+R_t)/(1+d))}{\ln((1+u)/(1+d))}$$

So $Y_T = \sum_{t=1}^{T} X_t$. We know that R_t can only take two values: u or d. Let us evaluate X_t for both cases:

   - If $R_t = u$ (probability q): $X_t = \frac{\ln((1+u)/(1+d))}{\ln((1+u)/(1+d))} = 1$.
   - If $R_t = d$ (probability $1-q$): $X_t = \frac{\ln((1+d)/(1+d))}{\ln((1+u)/(1+d))} = \frac{\ln(1)}{\ln((1+u)/(1+d))} = 0$.

5. **Conclusion:** Each X_t is a Bernoulli random variable with parameter q. Since the returns $R_1, \dots, R_T$ are independent and identically distributed (i.i.d.) under Q, the sum $Y_T = \sum_{t=1}^{T} X_t$ represents the sum of T i.i.d. Bernoulli random variables. Therefore, Y_T follows a **Binomial distribution** with parameters T and q: $Y_T \sim \mathcal{B}(T, q)$.

Question 2: Applying the Central Limit Theorem (CLT)

Goal: Provide an approximation of the law of S_T as $T \to \infty$.

Step-by-step Solution:

1. **Formulate the CLT:** Since Y_T is a sum of T i.i.d. random variables with mean $E_Q[X_t] = q$ and variance $Var_Q(X_t) = q(1-q)$, its total mean is Tq and total variance is $Tq(1-q)$. The Central Limit Theorem states that as $T \to \infty$, the standardized version of Y_T converges in distribution to a standard normal random variable $G \sim \mathcal{N}(0,1)$:
$$\frac{Y_T - Tq}{\sqrt{Tq(1-q)}} \approx G$$

2. **Isolate Y_T:** Rearrange the approximation to express Y_T in terms of G.
$$Y_T \approx \sqrt{Tq(1-q)}G + Tq$$

3. **Substitute Y_T back into the original definition:** Recall our definition of Y_T from Question 1.
$$\frac{\ln(S_T/(S_0(1+d)^T))}{\ln((1+u)/(1+d))} \approx \sqrt{Tq(1-q)}G + Tq$$

4. **Solve for S_T:** Now, we use algebra to isolate S_T. Multiply both sides by the denominator:
$$\ln\left(\frac{S_T}{S_0(1+d)^T}\right) \approx \ln\left(\frac{1+u}{1+d}\right)\left(\sqrt{Tq(1-q)}G + Tq\right)$$
Take the exponential of both sides:
$$\frac{S_T}{S_0(1+d)^T} \approx \exp\left[\ln\left(\frac{1+u}{1+d}\right)\left(\sqrt{Tq(1-q)}G + Tq\right)\right]$$
Using the property $\exp(a \cdot b) = (\exp(a))^b = (e^{\ln(x)})^b = x^b$, we can rewrite the right side:
$$\frac{S_T}{S_0(1+d)^T} \approx \left(\frac{1+u}{1+d}\right)^{\sqrt{Tq(1-q)}G+Tq}$$
Finally, multiply by $S_0(1+d)^T$ to get the approximation for S_T:
$$S_T \approx S_0(1+d)^T\left(\frac{1+u}{1+d}\right)^{\sqrt{Tq(1-q)}G+Tq}$$
This shows that in the limit, the asset price S_T follows a log-normal distribution, which is the foundational assumption of the Black-Scholes model.

Exercise 3: Pricing of Asian option

Context and Definitions

An Asian option is a path-dependent derivative where the payoff depends on the average price of the underlying asset over a certain period. Here, we consider a call option on the average with the payoff:

$$H_a := (A_T - K)^+$$

where $K > 0$ is the strike price, and $(A_t)_{0 \leq t \leq T}$ is the running average process defined as:

$$A_t := \frac{1}{t+1} \sum_{i=0}^{t} S_i$$

This average includes the initial price S_0 up to the current price S_t, meaning there are $t+1$ terms in the sum.

Question: Recursive Pricing Formula for an Asian Option

Goal: Let $v_t := E_Q[H_a/(1+r)^{T-t} \mid \mathcal{F}_t]$. Prove that v_t can be written as a function of the current asset price and the current average, $v_t = g_t(S_t, A_t)$, and find the recursive relationship g_{t-1}.

Step-by-step Solution: We will prove this using backward induction.

1. **Terminal Condition (Base Case $t = T$):** At maturity T, the value of the option v_T is simply the payoff itself, because there is no discounting needed (time to maturity is zero).

   $$v_T = H_a = (A_T - K)^+$$

   This payoff only depends on the final average A_T. Thus, we can define our base function as:

   $$g_T(s, a) := (a - K)^+$$

   The property holds for $t = T$.

2. **The Running Average Recursion (Crucial Step):** Before tackling the induction step, we need to express the average at time t (A_t) in terms of the average at time $t-1$ (A_{t-1}) and the new asset price S_t.

   $$\begin{aligned} A_t &= \frac{1}{t+1} \sum_{i=0}^{t} S_i \\ &= \frac{1}{t+1} \left(\sum_{i=0}^{t-1} S_i + S_t \right) \end{aligned}$$

   Notice that by definition, $A_{t-1} = \frac{1}{t} \sum_{i=0}^{t-1} S_i$, which implies that $\sum_{i=0}^{t-1} S_i = tA_{t-1}$. Substituting this back gives us the fundamental recursive update for the average:

   $$A_t = \frac{tA_{t-1} + S_t}{t+1}$$

3. **Backward Induction Step (Assuming true for t, proving for $t-1$):** Assume that at time t, the option price is given by $v_t = g_t(S_t, A_t)$. We want to find v_{t-1}. Using the Tower Property of conditional expectations, we step back one period:

   $$v_{t-1} = \frac{1}{1+r} E_Q[v_t \mid \mathcal{F}_{t-1}]$$

Substitute our induction hypothesis $v_t = g_t(S_t, A_t)$:

$$v_{t-1} = \frac{1}{1+r} E_Q[g_t(S_t, A_t) \mid \mathcal{F}_{t-1}]$$

Now, express S_t and A_t in terms of variables known at time $t-1$ and the independent return R_t. We know $S_t = S_{t-1}(1+R_t)$. Using the average recursion from Step 2:

$$A_t = \frac{tA_{t-1} + S_{t-1}(1+R_t)}{t+1}$$

Substitute these into the expectation:

$$v_{t-1} = \frac{1}{1+r} E_Q \left[g_t \left(S_{t-1}(1+R_t), \frac{tA_{t-1} + S_{t-1}(1+R_t)}{t+1} \right) \mid \mathcal{F}_{t-1} \right]$$

4. **Applying the Markov Property:** At time $t-1$, both S_{t-1} and A_{t-1} are completely known ($\mathcal{F}_{t-1}$-measurable). The only source of randomness is the return R_t, which is independent of the past filtration $\mathcal{F}_{t-1}$ under the risk-neutral measure Q. Therefore, the conditional expectation simply evaluates the function over the distribution of R_t. This proves that v_{t-1} is a deterministic function of S_{t-1} and A_{t-1}, which we define as g_{t-1}:

$$g_{t-1}(s, a) := \frac{1}{1+r} E_Q \left[g_t \left(s(1+R_t), \frac{ta + s(1+R_t)}{t+1} \right) \right]$$

This completes the proof. Note: To compute this expectation in practice, you sum over the two possible states for R_t (up with probability q, down with probability $1-q$) just like in standard European & American option pricing.